

Cramer's Rule

$$AX = B$$

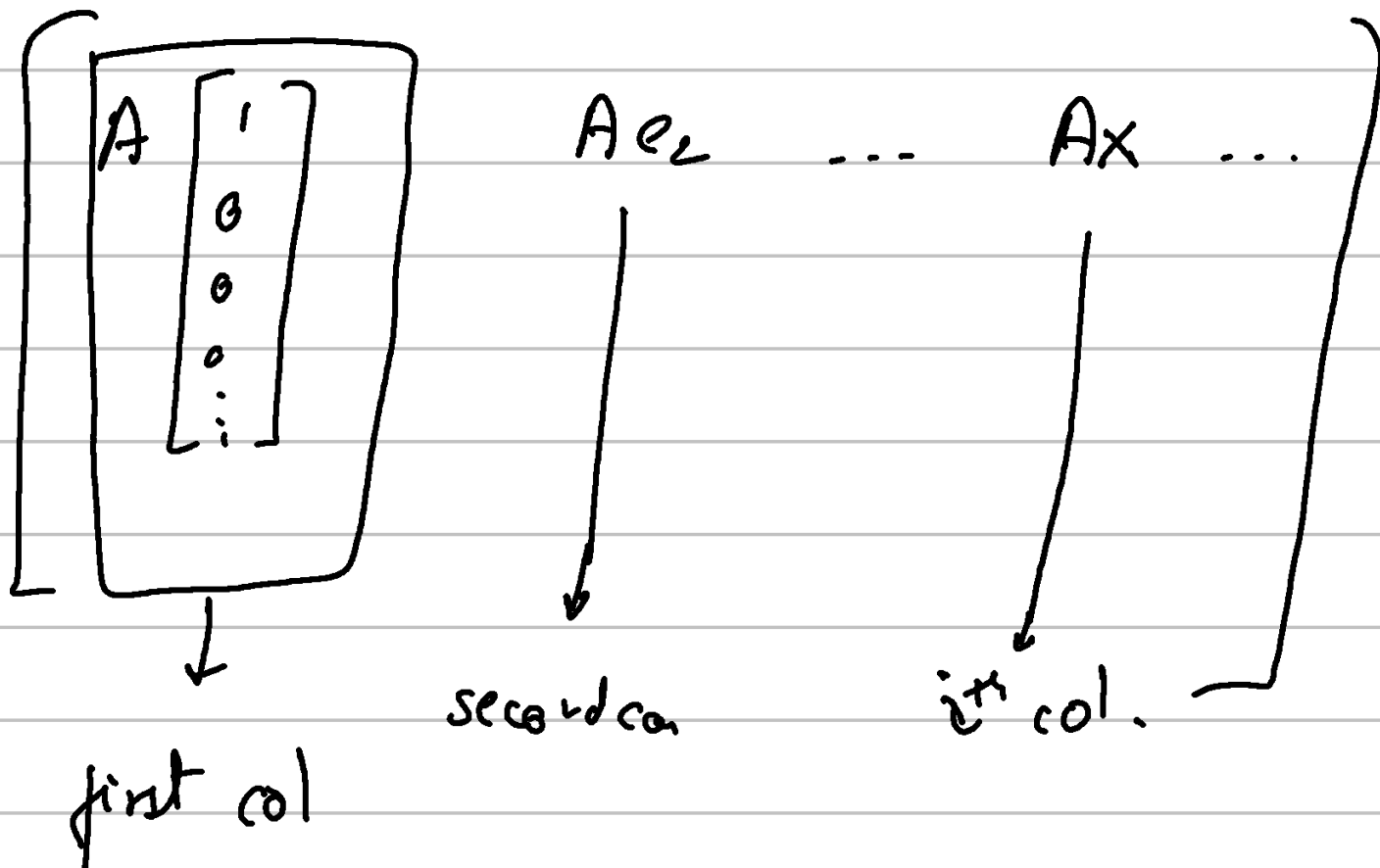
$$x_i = \frac{\det(A_i)}{\det(A)}$$

$A_i \rightarrow i^{\text{th}}$ column replaced by B

$\nearrow$ the solver

$$X_9 = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & & & \\ & & & & \\ 0 & & \boxed{}^x & & 0 \\ & & & & 1 \end{bmatrix}$$

$$AX_9 = \begin{bmatrix} \boxed{a_1} & \boxed{a_2} & \dots \end{bmatrix} \begin{bmatrix} 1 & 0 & \boxed{}^x \\ 0 & 1 & \vdots \\ 0 & 0 & \vdots \\ 0 & 0 & \vdots \\ \vdots & \vdots & \vdots \end{bmatrix}$$



$$= \left[\begin{array}{c|c|c|c|c} | & | & & | & | \\ a_1 & a_2 & \dots & b & \dots \\ | & | & & | & | \end{array} \right]$$

i -th col

$$= \underline{\underline{A_{\circ i}}}$$

$$A_i = A X_i$$

$$\det(A_i) = \det(A) \cdot \det(X_i)$$

$$\det(X_i) = 1 \cdot 1 \cdot 1 \cdot x_i \cdot 1 \cdot 1 = x_i$$

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Eigenvalues

$\Rightarrow$ For $v \neq 0$

$$Av = \lambda v$$

$(v, \lambda) \rightarrow$ eigen pair
 $\downarrow$
 eigenvector $\rightarrow$ eigen value.

⇒ Prove that roots of polynomial $\det(A - \lambda I) = 0$ are the eigenvalues.

$$Av = \lambda v$$

$$\begin{aligned}\Rightarrow Av - \lambda v &= 0 \\ (A - \lambda I)v &= 0\end{aligned}$$

$$v \in \text{Null}(A - \lambda I)$$

$$\text{Since } v \neq 0$$

$$\Rightarrow \dim(\text{Null}(A - \lambda I)) > 0$$

$$\Rightarrow A - \lambda I \text{ is not invertible.}$$

$$\Rightarrow \det(A - \lambda I) = 0$$

$\Rightarrow$ A is called diagonalizable if we can write it as

$$A = P D P^{-1}$$

$$A^k = P D^k P^{-1}$$

$\Rightarrow$ Prove that being diagonalizable is equivalent to the fact that the eigenvectors of A are a basis of $\mathbb{R}^n$

$$A P = \begin{bmatrix} | & | & & | \\ Av_1 & Av_2 & \dots & Av_n \\ | & | & & | \end{bmatrix}$$

$$PD = \begin{bmatrix} 1 & & & \\ \lambda_1 v_1 & \lambda_2 v_2 & \dots & \lambda_n v_n \\ 1 & & & \end{bmatrix}$$

- Hence cols of P are eigenvectors.
- Values on diagonal of D are eigenvalues.

Symmetric Matrices

Lemma

⇒ A real symmetric matrix has only real eigenvalues and admits a real eigenvector.

$$\boxed{\overline{z z'} = \bar{z} \bar{z'}} \text{ nice property.}$$

$$z^* = \overline{z^T}$$

$$\text{Now, } z^* A z = \lambda z^* z$$

$$z^* A z = z^* A^* z$$

because A is real & symmetric
so, $A = A^*$

$$z^* A z = z^* A^* z$$

$$z^* A z = (z^* A^* z)^*$$

$$\text{So, } z^* A^* z = (z^* A z)^*$$

Hence $z^* A z$ is a real and symmetric matrix.

$$\begin{aligned} z^* A z &= \lambda z^* z \\ &= \lambda \bar{z}^T z \\ &= \lambda \langle \bar{z}, z \rangle \\ \underbrace{z^* A z}_{\text{real}} &= \lambda \underbrace{\|z\|^2}_{\text{real}} \end{aligned}$$

Hence λ has to be real.